



XMBA 200S: Data Analysis for Management

Chapter 17: Comparisons and Experiments

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Course overview:

Part 2: Making decisions based on inference

- Given **limited data**, managers need to choose how to act
 - Typically, you'll have some benchmark in mind, or two (or more) groups for comparison
 - But how do you know any difference isn't due to "noise"?
 - What is the likelihood of making a mistake?
- What we need are **statistical tests** that provide us with some clarity
- This class: tests comparing two samples

Course overview:

Part 3: Experiments and causality

- In many comparisons you make, you want to attribute differences to some **causal factor**
- Randomized Controlled Trials (RCTs) are a uniquely powerful method of testing for causal effects
 - Non-experimental comparisons may suffer from “lurking variables”
 - Observational relationships are rarely causal.
 - Correlation is not causation!

What to take away from this class

- How to **conduct** statistical tests comparing two samples
- How to **interpret** comparisons, keeping in mind the “ideal experiment”, and thinking about how close you are to it

Outline

- Two sample tests
- Introduction to experiments
- Hawthorne case

Education Technology

Great Minds is a company focused on curriculum development for K-8 education. Movement within education to standardize curriculum to...

- reduce teaching disparities across schools
- lower the costs of teacher turnover / professional development, and
- reap benefits of specialization.

Curriculum is licensed to a school (or school district or charter network) for a licensing fee.

Whether or not a school or district adopts the curriculum depends on how successful it is in boosting student achievement.

Is it more effective than the status quo? Is the new curriculum effective enough for a school district to pay the licensing fees?



Education

Evaluation of a New Curriculum

- The population of interest is all schools who would use Great Mind's curriculum.
- We would like to compare two means: mean of the “treatment” group and mean of “control” group.
- *Treatment Group*: Let μ_G denote the mean test score in the population for schools who receive the Great Mind's curriculum.
- *Control Group*: Let μ_C denote the mean test score in the population for schools who use the status quo teaching / instruction.

Education Technology

Comparing Two Learning Models

The school district only believes it makes sense to license / adopt the new curriculum if it will lead to an improvement in state test scores of more than 2 points on average.

State the hypotheses as:

Data for Comparisons

- Data used to compare two groups typically arise in one of three ways:
 1. Compare two sets of observations.
 2. Obtain random samples from two populations.
 3. Run an experiment that isolates a specific cause

Experiments

Basic Terminology

Experiment – procedure that uses randomization to produce data that reveal causation

Treatment – something done to an experiment participant that might have an effect; typically this is described using one variable that takes on different values, (e.g. 1 if treated, 0 otherwise).

Response – where we look to see if the treatment had an effect; typically this is described using one or more response variables

Data for Comparisons

Experimentation

In the ideal experiment, the experimenter:

1. Selects a random sample from a population
 - Crucial for external validity (can you generalize from your sample?)
2. Assigns subjects at random to treatments (and control)
 - Crucial for internal validity (have you actually identified the causal effect of a specific treatment?)
3. Compares the response of subjects between treatments

Why randomization is valued: eliminating confounding factors

Confounding

- The only difference between treatment and control on average should be the treatment that they receive
 - Randomization means other factors should not be systematically associated with specific treatments
- If there may be additional differences, we say there are potential confounding factors
- **Selection** into particular treatments is a primary concern with non-experimental, “observational” studies

Confounding?

HEALTH

371 COMMENTS

Meat Is Linked to Higher Cancer Risk, W.H.O. Report Finds

By ANAHAD O'CONNOR

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An international panel of experts convened by the [World Health Organization](#) concluded Monday that eating processed meat like hot dogs, ham and bacon raises the risk of [colon cancer](#) and that consuming other red meats “probably” raises the risk as well. But the increase in risk is so slight that experts said most people should not be overly worried about it.

The panel did not offer specific guidelines on red meat consumption. But its conclusions add support to recommendations made by other scientific groups like the federal government’s dietary guidelines advisory committee, which has long discouraged the consumption of red and processed meat. And the report could also influence health agencies such as the European Food and Safety Commission.



Bacon cooks in a frying pan. A W.H.O. report said eating processed meats can raise the risk in humans of getting colon cancer. Joe Raedle/Getty Images

Confounding?

Q&A on the carcinogenicity of the consumption of red meat and processed meat

Q. Red meat was classified as Group 2A, *probably carcinogenic to humans*. What does this mean exactly?

A. In the case of red meat, the classification is based on *limited evidence* from epidemiological studies showing positive associations between eating red meat and developing colorectal cancer as well as *strong* mechanistic evidence.

Limited evidence means that a positive association has been observed between exposure to the agent and cancer but that other explanations for the observations (technically termed chance, bias, or confounding) could not be ruled out.

http://www.iarc.fr/en/media-centre/iarcnews/pdf/Monographs-Q&A_Vol114.pdf

Why randomization is valued: eliminating confounding factors

Examples of observational associations (with confounding):

- People who receive heart transplants are likely to have shorter life expectancies than people who do not
 - Are heart transplants a bad thing?
- Kids who enroll in a test prep program do better on exams
 - Is that all coming from the test prep?
- Firms and/or sports teams that change managers often see improved performance
 - Do new managers typically do such a great job?

The challenge of identifying causal effects

Generally, after observing a pattern in data, you need to ask:

1. What causal effect is suggested?
2. What alternative stories are there for the observed pattern?
3. How might you tell which hypothesis is most plausible?
 - More data needed?
 - Additional tests?
 - An experiment...

Experiments

Benefits of Randomized Control Trials

1. Randomized Control Trials have no bias.
 - Though, be careful of “attrition”
2. Random assignment eliminates lurking variables
3. Randomized control trials are transparent
 - Simple to explain to non-experts and highly credible.

Costs of Experimentation

1. Randomized Control Trials may be expensive to run
2. Randomization is not an ethically viable method in certain situations (in 2014 Facebook revealed news feed experiment to control user's emotions)
3. Randomized control trials require replication
 - Without multiple replications we may still have Type I or Type II errors (e.g. publication bias against null results)

Comparing Sample Means

Comparing Responses

- With data from a well-designed experiment, it is easy to compare sample means from the treatment and control groups.
- Or, to compare the difference to some break-even level.
 - Break-even could come from profitability calculations
- And for assessing whether these differences are statistically significant, the most used test is the two-sample t-test.
- Later on, we'll show you how to do this with a regression!

Comparing Sample Means

Two-Sample t-Statistic (by hand, in case of Apocalypse)

The formula for the two-sample t-statistic is,

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - D_0}{se(\bar{x}_1 - \bar{x}_2)}$$

Where D_0 is the break-even value (specified by the null hypothesis)

Note: Degrees of freedom are approximated using the rule of thumb $n_1 + n_2 - 2$. For more accuracy, rely on software to calculate degrees of freedom and p -values.

Comparing Sample Means

Gory Details (software will do this for you)

Population parameters	μ_1, μ_2
Null hypothesis	$H_0: \mu_1 - \mu_2 \leq D_0$
Alternative hypothesis	$H_a: \mu_1 - \mu_2 > D_0$
Sample statistic	$\bar{x}_1 - \bar{x}_2$
Estimated standard error	$\text{se}(\bar{X}_1 - \bar{X}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
Test statistic	$t = (\bar{X}_1 - \bar{X}_2 - D_0) / \text{se}(\bar{X}_1 - \bar{X}_2)$
Reject H_0 if	$p\text{-value} < \alpha$ or $t > t_\alpha$

Comparing Sample Proportions

Gory Details (software will do this for you)

Population parameters	p_1, p_2
Null hypothesis	$H_0: p_1 - p_2 \leq D_0$
Alternative hypothesis	$H_a: p_1 - p_2 > D_0$
Sample statistic	$\hat{p}_1 - \hat{p}_2$
Estimated standard error	$\text{se}(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$
z-statistic	$z = (\hat{p}_1 - \hat{p}_2 - D_0) / \text{se}(\hat{p}_1 - \hat{p}_2)$
Reject H_0 if	$p\text{-value} < \alpha \text{ or } z > z_\alpha$

A Statistical Test of Interest

What changes with two-sample tests?

1. The statistic of interest: no longer just one statistic from a sample, now it is a difference between two statistics
2. The comparison: now compare to a “break even difference”, rather than a population mean or population proportion
3. SE's: now remember to use the formula for SE(difference), rather than the simple formula for SE
4. For means, degrees of freedom (t-tables!): the sum of the two sample sizes minus 2

A Curriculum Experiment

Experiment Design

Before implementing the curriculum district wide, the New York City Department of Education decides to pilot the curriculum using an experiment.

The district randomly selects 63 elementary schools out of the 700+ in city.

- Treatment: 33 randomly chosen to use Great Mind's curriculum
- Control: remaining 30 teach using conventional methods.

Experimental Pitfalls

1. Non-compliance

- Experimental participants do not *comply* with treatment
- Analysts often report “Intention to Treat” (ITT) estimates
 - Average difference between people assigned to treatment versus people assigned to control
- Affects interpretation of results, but can still deliver causal relationships if comparing all treatment obs. to all control obs.

2. Attrition

- Experimental participants *drop out of study*
- Experimental comparisons may now be *biased*, and tests may no longer be causal

3. Testing too many or not “pre-specified” outcomes

- With significance level = 5%,
- Type I error rate of 20 outcomes = $1 - 0.95^{20} = 0.64$

Ask/watch out for these issues when interpreting experiments!

A Curriculum Experiment

Descriptive Statistics

	Number	Mean	Std. Dev	Skewness	Kurtosis
Great Minds Curriculum	$n_G = 33$	$\bar{x}_G = 15.42$	$s_G = 14.37$	-0.052	0.100
Conventional	$n_C = 30$	$\bar{x}_C = 7.00$	$s_C = 12.36$	0.342	-0.565

Mean test scores with Great Minds are larger than mean test scores with conventional instruction.

- Is the difference statistically significant?
- Is the difference large enough to make the school district adopt the new curriculum?

A Curriculum Experiment

Calculating the t-Statistic

First calculate the standard error of the difference between the two means,

$$se(\bar{x}_G - \bar{x}_C) = \sqrt{\frac{s_G^2}{n_G} + \frac{s_C^2}{n_C}} = \sqrt{\frac{14.37^2}{33} + \frac{12.36^2}{30}} = 3.369$$

Then calculate the t-statistic,

$$t =$$

With degrees of freedom equal to

$$n_1 + n_2 - 2 = 61.$$

On the t-table this corresponds to a p -value on table between

Thus, we _____ H_0 at a 5% significance level.

What Does Claude Say?

	Number	Mean	Std. Dev	Skewness	Kurtosis
Great Minds Curriculum	$n_G = 33$	$\bar{x}_G = 15.42$	$s_G = 14.37$	-0.052	0.100
Conventional	$n_C = 30$	$\bar{x}_C = 7.00$	$s_C = 12.36$	0.342	-0.565

test for the difference in means compared to a one sided null of 2

Running a one-sided Welch test against the threshold difference. >

Reject H_0 — the difference in means exceeds 2.

Taking $H_0: \mu_G - \mu_C \leq 2$ vs $H_1: \mu_G - \mu_C > 2$ (two-sample t-test, unequal variances):

- Observed difference: $15.42 - 7.00 = \mathbf{8.42}$
- $SE = \sqrt{(14.37^2/33 + 12.36^2/30)} = \sqrt{(6.257 + 5.092)} = \mathbf{3.369}$
- $t = (8.42 - 2)/3.369 = \mathbf{1.906}$
- Welch $df \approx 61$; critical $t_{0.05} \approx \mathbf{1.671}$
- One-tailed $p \approx \mathbf{0.031}$

Since $1.906 > 1.671$ ($p \approx 0.03 < 0.05$), you reject the null that the Great Minds advantage is at most 2 points.

A Curriculum Experiment

The experiment shows that the average test score with the Great Minds curriculum exceeds the average test score with the conventional instruction by more than two points.

What about external validity?

- e.g., this experiment comes from a subset of schools in NYC
- Do we think this would generalize to other schools in NYC?
Elsewhere?

Other Concerns?

Confidence Interval for the Difference

95% Confidence Intervals for μ_G and μ_C

Great Minds		Conventional
15.42	Mean	7.01
14.37	Std Dev	12.36
2.50	Std Err Mean	2.26
20.52	Upper 95% Mean	11.62
10.33	Lower 95% Mean	2.39
33	<i>n</i>	30

Confidence Interval for the Difference

95% Confidence Interval for $\mu_1 - \mu_2$

The $100(1 - \alpha)\%$ two-sample confidence interval for the difference in means is

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\frac{\alpha}{2}} \times se(\bar{x}_1 - \bar{x}_2)$$

Confidence Interval for the Difference

- 95% Confidence Interval for $\mu_G - \mu_C$

Difference	8.42
Std. Error Difference	3.37
95% Confidence Interval	1.68 to 15.15

- Since the 95% confidence interval for $\mu_G - \mu_B$ does not include 0, the means are statistically significantly different
- Those using Great Mind's curriculum have between 1.7 and 15.2 point higher test scores than those using conventional instruction

Takeaways

Observational studies usually suffer from lurking variables.

Use experiments to discover causal relationships.

Randomized control trials ensure both internal and external validity.

Use two-sample t-tests to compare sample means and sample proportions.

Best Practices / Experimental Playbook

Pre-register hypotheses + Power Calculations
(which test/sample size/significance level)

Balance Table at Baseline: Checking randomization “worked”

Balance Table at Endline:

- Checking that participants at endline still look “balanced” or is there sample selection (non-random attrition). If selection, see previous lectures (bounds, weighting, etc...)

Other analysis/design considerations:

- Randomized encouragement not treatment
 - Estimate “intent to treat” differences or use instrumental variable techniques to estimate “treatment on the treated”
- Use tools to correct for multiple hypothesis testing

Beyond This Course: Theory of Experimentation

Bridging Data Science, Strategy, Economics... How large companies can act like startups:

- Advice: Focus on finding radical surprises.
- Why? A/B tests at Bing show most site tweaks have tiny benefits requiring large tests to identify (intuition: expensive with low return)
- Strategy: Instead, run more, small & less accurate tests: the top 2% of ideas are responsible for 74.8% of the gains (intuition: cheaper with high return)

A/B Testing with Fat Tails*

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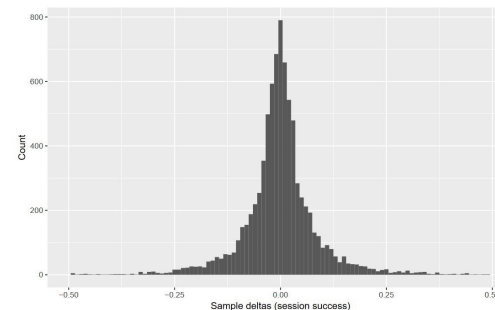


Figure 2: Distribution of measured deltas in session success

Notes: The figure displays a histogram of sample deltas, or signals, of session success, across all experiments.

Outline

- Two sample tests
- Introduction to experiments
- Hawthorne case

Hawthorne Experiment

Exhibit A: Mean Daily Productivity (Units per Day)

	Number of Observation	Mean	Standard Deviation	Skewness	Kurtosis
Normal Lighting	320	452.2	84.4	-.18	.09
Improved Lighting	220	472.4	85.8	.19	.10

Hawthorne Experiment

Exhibit B: Mean Daily Productivity by Department (Units per Day)

		Number of Observations	Mean	Standard Deviation	Skewness	Kurtosis
Department 1	Normal Lighting	60	512.2	74.4	-.28	.18
	Improved Lighting	120	503.4	81.0	.07	-.46
Department 2	Normal Lighting	130	439.8	79.6	-.28	.42
	Improved Lighting	50	434.4	80.6	.75	2.18
Department 3	Normal Lighting	130	436.9	81.8	-.15	-.06
	Improved Lighting	50	435.7	72.7	-.14	1.49

Hawthorne Effects: Psychology



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Hawthorne effect

From Wikipedia, the free encyclopedia

The **Hawthorne effect** (also referred to as the **observer effect**) is a type of [reactivity](#) in which individuals modify or improve an aspect of their behavior in response to their awareness of being observed.^{[1][2]} The original research at the [Hawthorne Works](#) in Cicero Illinois on lighting changes and work structure changes such as working hours and break times were originally interpreted by Elton Mayo and others to mean that

paying attention to overall worker needs would improve productivity. Later interpretations such as that done by Landsberger suggested that the novelty of being research subjects and the increased attention from such could lead to temporary increases in workers' productivity. This interpretation was dubbed "the Hawthorne effect."



Aerial view of the Hawthorne Works, ca. 1925.



Experimentation in Action

Credit Indemnity

- Credit Indemnity is one of the largest micro-lenders in South Africa; a highly profitable lender with 20 years of experience
- Offers short-term, high-interest, uncollateralized credit for people without other access to credit
- The typical loan is about \$150, about a third of the borrower's monthly income, most of the loans last about 4 months
- For many years has used experiments to evaluate the effectiveness of its marketing efforts.

Experiment

- Credit Indemnity recently performed an experiment in which they sent credit offers by mail to 50,000+ households
- Credit Indemnity randomly varied the offers in several ways including different interest rates.
- The parameter of interest is the proportion of individuals who applied for a loan in the month following the offer.

Variable	Obs	Mean	Std. Dev.	Min	Max
applied	53194	.0850096	.2788986	0	1

Business Question

How do interest rates and other factors affect the demand for short-term loans from Credit Indemnity?

- Firms spend billions each year on advertising, but there is surprisingly little quantitative evidence on its effectiveness.
- In part, this is because it difficult to evaluate the effectiveness of advertising using observational studies.
- What works? What doesn't work?

Treatment 1: Interest Rates

Rates varied from 3.25% to 11.75% per month.

Variable	Obs	Mean	Std. Dev.	Min	Max
rate	53194	7.929017	2.421657	3.25	11.75

“Congratulations! As a valued client, you are now eligible for a special interest rate on your next cash loan from Credit Indemnity. Enjoy low monthly repayments with this offer!”

Treatment 2: Including a Photo

80% of the letters included a photo.

Both female and male photos were used.

The other 20% had no photo.

Variable	Obs	Mean	Std. Dev.	Min	Max
photo_female	53194	.398936	.4896842	0	1
photo_male	53194	.3991428	.4897268	0	1



Treatment 3: Number of Examples

Either one or four hypothetical loan amounts were provided.

Loan Amount	Loan Term	Monthly Repayment
R1000	4 Months	R290

Loan Amount	Loan Term	Monthly Repayment
R500	4 Months	R145
R1000	4 Months	R290
R2000	4 Months	R580
R4000	4 Months	R1160

All offers also said, “Loans available in other amounts.”

Variable	Obs	Mean	Std. Dev.	Min	Max
one_example	53194	.496447	.4999921	0	1

Treatment 4: Report the Interest Rate

In some offers the interest rate was shown explicitly and in other offers it was not.

Interest Rate	Loan Amount	Loan Term	Monthly Repayment
3.99%	R1000	4 Months	R290

Loan Amount	Loan Term	Monthly Repayment
R1000	4 Months	R290

Variable	Obs	Mean	Std. Dev.	Min	Max
rate_shown	53194	.799921	.400063	0	1

Treatment 5: Suggested Activities

All the offers made some statement providing a suggestion of what you can do with the money.

Alternatives included:

- (1) “You can use this cash to pay for school, or for anything else you want.”
- (2) “You can use this cash to pay off a more expensive debt, or for anything else you want.”
- (3) “You can use this cash to repair your home, or for anything else you want.”
- (4) “You can use this cash to buy an appliance, or for anything else you want.”
- (5) “You can use this cash for anything you want.”

Variable	Obs	Mean	Std. Dev.	Min	Max
sug_school	53194	.2008309	.4006256	0	1
sug_debt	53194	.2007557	.4005694	0	1
sug_home	53194	.1988006	.3991014	0	1
sug_applian	53194	.1992142	.3994132	0	1

Which do you think matters?

Treatment 1: Interest Rate

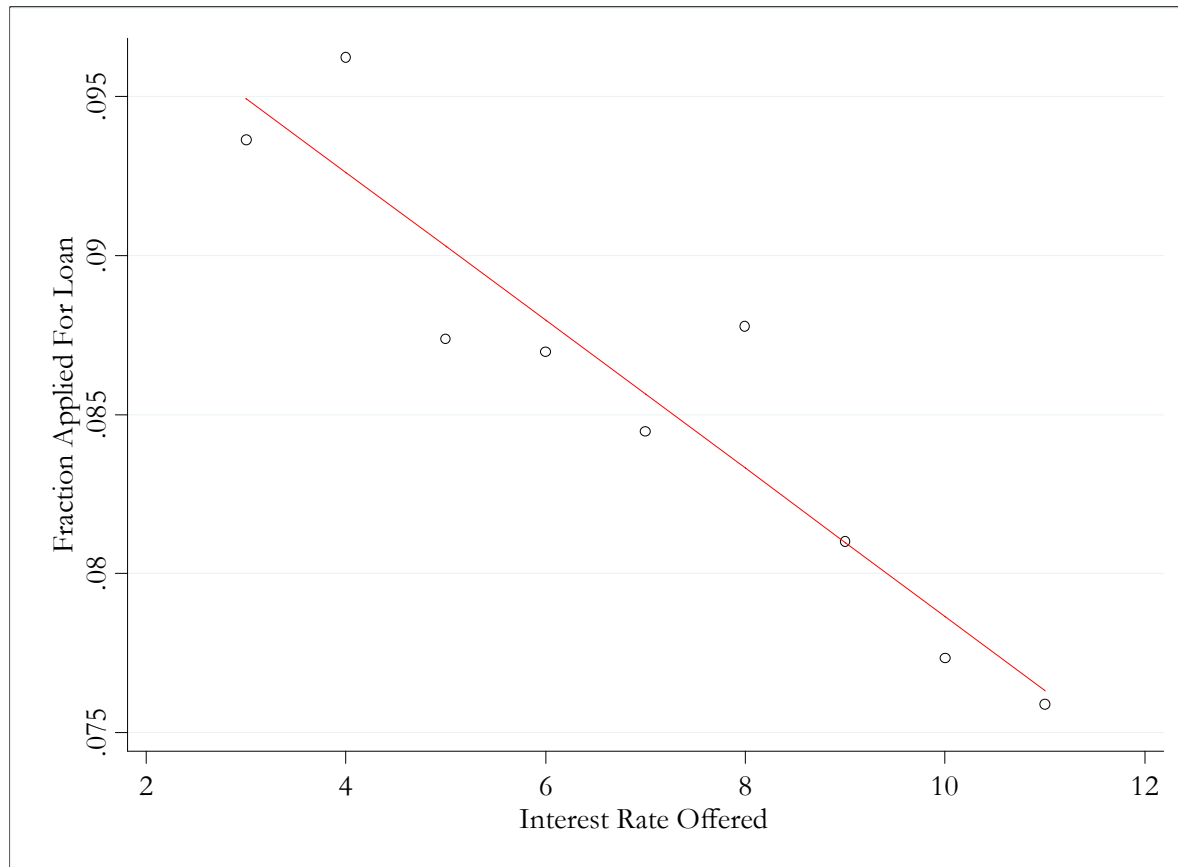
Treatment 2: Including Photos

Treatment 3: Number of Examples

Treatment 4: Include the Interest Rate

Treatment 5: Suggested Activities

Treatment 1: Interest Rates



- Probability of applying falls $3/10^{\text{th}}$ of a percentage point for every 100 basis point increase in monthly interest rate

Treatment 2: Including Photos

Including the photo of a woman increased applications by 0.5 percentage points off of a mean of about 8%.

- This is equivalent to a 2 percentage point reduction in the interest rate.

The effect of including the photo of a man is small and statistically insignificant.



Treatment 3: Number of Examples

Loan Amount	Loan Term	Monthly Repayment
R1000	4 Months	R290

Loan Amount	Loan Term	Monthly Repayment
R500	4 Months	R145
R1000	4 Months	R290
R2000	4 Months	R580
R4000	4 Months	R1160

Including only a single example increased loan applications by 0.6 percentage points.

Treatments 4: Include the Interest Rate

Interest Rate	Loan Amount	Loan Term	Monthly Repayment
3.99%	R1000	4 Months	R290

Loan Amount	Loan Term	Monthly Repayment
R1000	4 Months	R290

There is no evidence that showing the interest rate explicitly had any observable impact on loan applications.

Treatment 5: No Suggested Activity

The most successful suggested activity is no suggested activity.

“You can use this cash for anything you want.”

None of the other specific suggestions had any impact.

Results Summary

Thus the most effective interventions seem to be:

- 1) Lower Interest Rate
- 2) Including the Photo of a Woman
- 3) Including Only One Example
- 4) No Suggested Activity

What, if any, broader lessons can be drawn here?

What else would you try if you were Credit Indemnity?

Implications

- Overall, the results show that seemingly non-informative advertising plays a large role in real consumer decisions.
- This suggests that it is difficult to predict *ex ante* which specific content features will significantly impact demand.
- Experimentation can offer a low-cost approach for discovering what matters, regardless of what you are selling.